Formally Verified Derivation of the Standard Model Gauge Group from a Common Algebraic Framework

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March 18, 2026

Abstract

We present a formally verified framework that derives the Standard Model gauge group SU(3)×SU(2)×U(1), its complete one-generation fermion content, and all hypercharge ratios from seven explicit inputs: five structural primitives (source functional, Lie algebra, Lorentzian metric, linear composition, perturbation dimension ≥ 2), one physical hypothesis (energy boundedness), and one selection principle (minimality of the fermion count). The derivation chain is: chirality (from the source/dressing decomposition) requires complex representations, which restricts the gauge algebra to SU(N) type; the K/P structure forces two factors (one chiral, one vector-like) that must be non-isomorphic as represented algebras (proven: a surjective equivalence would transport the vector-like property, contradicting chirality); the representation structure is derived from color parity and chirality (the only chiral color-parity-satisfying structure is $(N_c, N_w) + N_w \times (\bar{N}_c, 1) + (1, N_w) + (1, 1)$); charge determinacy uniquely selects SU(2) as the weak factor ($N_w = 2$ is the only value giving fully determined hypercharges); distinctness and minimality select SU(3) as the color factor; U(1) arises independently from the dressing sector; and anomaly cancellation — via a universal cubic factorization $6N_c(N_c - r - 1)(N_c + r + 1) = 0$ valid for any SU(N_c) — uniquely determines all hypercharge ratios as $(1, -4, 2, -3, 6) \cdot y_Q$.

The framework additionally derives: the Einstein field equation as the unique second-order gravitational dynamics in 4D (with the second-order condition from stability via the Ostrogradsky instability theorem); full nonabelian Yang–Mills structure; the propagation rule e^{ikL} from the source functional's linearity (yielding an unbroken chain from primitives to interference); the Born-type observable; the Higgs potential as the unique quartic rotation-invariant on the K/P space; symmetry breaking from decoherence; the graviton as a traceless P-sector defect with 2 polarizations in $d = 3$; minimal coupling; a strong CP resolution; $N_g = 3$ from the holomorphic sections of $\mathcal{O}(1)$ on the gauge orbit fiber $\mathbb{CP}^2$; and $N_g \geq 3$ independently from CP violation. A zero-parameter prediction gives $\alpha_2(M_Z) \approx 0.036$ (measured: 0.034, 7% error). The algebraic core is formally verified in Lean 4 with zero custom axioms and zero `sorry` statements.

1 Introduction

We identify a small parent architecture and prove that gravity, gauge structure, matter content, and quantum-style amplitudes are uniquely selected or conditionally induced

from it. The algebraic core is formally verified in Lean 4; spacetime realizations are computationally verified in Python.

The headline result: the Standard Model gauge group, its matter content, and the number of generations are *derived*, not assumed. From seven inputs — five primitives, energy boundedness, and minimality — the complete chain from abstract algebra to SU(3)×SU(2)×U(1) with 15 fermions per generation, $N_g = 3$ generations, and unique hypercharges is machine-checked in Lean 4 with zero `sorry` statements and zero custom axioms.

The full derivation tree:

Input	$\longrightarrow$	Derived structure
Source functional φ	$\longrightarrow$	K/P split $\to$ chirality $\to$ complex reps $\to$ SU(N) type $\to$ two factors $\to$ rep structure forced (color parity) $\to$ chiral $\not\cong$ vector-like $\to$ SU(3) (color) $\to$ fiber $\mathcal{O}(1)$ sections $\to N_g = 3$ $\to$ U(1) from dressing $\to$ SSB from decoherence
Lie algebra	$\longrightarrow$	Yang–Mills, Killing form $\to$ unique action
Lorentzian metric	$\longrightarrow$	Riemann, Bianchi, Lovelock $\to$ Einstein $+\Lambda$
Linear composition	$\longrightarrow$	Charges, anomaly cancellation $\to$ hypercharges
$\dim(V) \geq 2$	$\longrightarrow$	$z = Q + iP$, $\lvert z\rvert^2$, e^{ikL}, interference
Stability	$\longrightarrow$	Compactness, Ostrogradsky $\to$ 2nd order, Higgs potential
Minimality	$\longrightarrow$	Charge determinacy $\to$ SU(2), distinctness $\to$ SU(3), 15 fermions

1.1 Formal verification

Every algebraic theorem is formally verified in Lean 4 [1] with the Mathlib library (`https://github.com/tomdif/unifiedtheory`). Zero errors, zero `sorry`. Axiom footprint: {`propext`, `Classical.choice`, `Quot.sound`}.

1.2 Relation to prior work

The idea that anomaly cancellation constrains the Standard Model is classical: Adler [2] and Bell–Jackiw [3] established the chiral anomaly; Bouchiat, Iliopoulos, and Meyer [4] first showed that anomaly cancellation restricts the fermion charges. The systematic classification of anomaly-free chiral gauge theories was advanced by Minahan, Ramond, and Warner [5]. Grand unified embeddings (Georgi–Glashow [6], Pati–Salam [7]) derive the SM charges from a larger group but introduce new inputs (the GUT group itself, its breaking pattern). Our contribution is different in kind: we derive the gauge group, representation content, *and* hypercharge ratios from a common algebraic framework with machine-checked proofs, and we identify the precise inputs required (seven, no more).

The formal verification methodology extends the tradition of Hales's Flyspeck project [16] and Gonthier's formalization of the Feit–Thompson theorem [17], applying computer-verified proof to theoretical physics for the first time at this scope.

2 Gravity

Theorem 1 (Renormalization rigidity). *The unique scale-invariant exponent is* $\alpha = d-1$. *In* 3+1*D:* $\alpha = 2$.

Theorem 2 (Contracted Bianchi identity). $\nabla^a G_{ab} = 0$ *(kinematic, all metrics).*

Theorem 3 (Complete 4D Lovelock uniqueness). *In the second-order natural class (second-order from Ostrogradsky [11], §9.1): the unique divergence-free rank-2 tensor is* $a\,G_{ab} + \Lambda\, g_{ab}$. *Lovelock [10] determines the* form*; the discrete substrate [9, 8] suggests the* magnitude $\Lambda \sim N^{-1/2} \sim 10^{-120}$.

Theorem 4 (Dimension selection [12, 13]). $d = 3$ *is the unique spatial dimension with stable orbits and Huygens propagation.*

Theorem 5 (Graviton). *A traceless metric perturbation* ($\varphi(h) = 0$) *satisfies* $\varphi(K(h)) = 0$ *by the bridge equation* $\varphi(K(v)) = \varphi(v)$*: the K-projection carries zero source charge. The graviton lives entirely in the P-sector. In* $d = 3$*: 2 polarizations.*

3 Derivation of the Standard Model Gauge Group

This section presents the central result: the complete derivation of SU(3)×SU(2)×U(1) from the framework's inputs.

Theorem 6 (Nonabelian Yang–Mills). *From structure constants with antisymmetry and Jacobi: the Killing form gives the unique gauge-invariant quadratic action;* $D^\mu F_{\mu\nu} = 0$ *is its stationarity condition; the Bianchi identity is fully proven (kinematic).*

Theorem 7 (Chirality from K/P). *Gauge invariance of* φ *+ K/P decomposition: the gauge group acts differently on source (K) and dressing (P) sectors. Chirality is derived, not postulated.*

Theorem 8 (Complex representations required). *Chirality requires left-handed and right-handed fermions to transform in inequivalent representations. This is only possible if the gauge algebra has* complex *representations (representations not isomorphic to their conjugates). Among compact simple Lie algebras, the classification of representation types is [15]:*

Algebra	*Fundamental rep type*	*Chiral?*
$\mathfrak{su}(N)$, $N \geq 3$	*Complex*	*Yes*
$\mathfrak{so}(4k+2)$	*Complex (spinor)*	*Yes*
$\mathfrak{e}_6$	*Complex*	*Yes*
$\mathfrak{su}(2)$, $\mathfrak{so}(2k+1)$, $\mathfrak{sp}(N)$	*Pseudoreal*	*No*
$\mathfrak{so}(4k)$ *(*$k \geq 2$*)*, $\mathfrak{e}_7$, $\mathfrak{e}_8$, $\mathfrak{f}_4$, $\mathfrak{g}_2$	*Real*	*No*

Minimality of the fundamental representation dimension selects SU(N): dim $= N$ *for SU(N), versus* dim $= 16$ *for SO(10) and* dim $= 27$ *for* E_6. *The chiral gauge algebra factors are therefore of SU(N) type.*

Theorem 9 (Two factors from K/P structure)**.** *The K/P decomposition requires two kinds of gauge action: chiral (different on K and P) and vector-like (same on both). A single simple factor provides only one kind. Two factors — one chiral (the weak group), one vector-like (the color group) — are the minimal structure accommodating both.*

Theorem 10 (Representation structure forced)**.** *The fermion representation content* $(N_c, N_w) + N_w \times (\bar{N}_c, 1) + (1, N_w) + (1, 1)$ *is* derived*, not assumed. Color parity (vanishing of the* $SU(N_c)^3$ *anomaly [2, 3]) requires equal positive and negative contributions. If both colored sectors carry* $SU(N_w)$ *multiplets, color parity forces equal multiplicities, making the theory vector-like (contradicting chirality). The unique chiral solution has one* (N_c, N_w) *multiplet and* N_w *copies of* $(\bar{N}_c, 1)$*, with the lepton sector* $(1, N_w) + (1, 1)$ *required for the mixed anomaly.*

Theorem 11 (Distinctness from chirality)**.** *The weak factor acts* chirally *on* $V = K \oplus P$*: its restriction to* K *is inequivalent to its restriction to* P *(from Theorem 7). The color factor acts* vector-like*: gauge invariance of the source functional* φ *forces the color action to commute with the K/P projection, so its restrictions to* K *and* P *are equivalent. An isomorphism* $\sigma : \mathfrak{g}_c \to \mathfrak{g}'$ *of represented algebras would satisfy* $\rho'(\sigma(x))|_K \cong \rho_c(x)|_K$ *and* $\rho'(\sigma(x))|_P \cong \rho_c(x)|_P$*. Since* $\rho_c(x)|_K \cong \rho_c(x)|_P$ *(vector-like) and* σ *is surjective, this yields* $\rho'(y)|_K \cong \rho'(y)|_P$ *for all* $y \in \mathfrak{g}'$*, contradicting chirality. Therefore* $G_c \ncong G'$*.*

Theorem 12 (U(1) from dressing)**.** *The complex structure* $z = Q + iP$ *forced by Frobenius carries a phase rotation* $z \mapsto e^{i\theta} z$*: a U(1) symmetry with a different geometric origin than the nonabelian factors.*

Theorem 13 (SU(2) from charge determinacy and minimality)**.** *The fermion count for a weak factor* $G' = \mathrm{SU}(N_w)$ *is* $7N_w + 1$ *(from color parity + lepton sector counting). For* $N_w = 1$ *the theory is vector-like (excluded by chirality). Among chiral values* $N_w \geq 2$*, the charge variable count is* $N_w + 3$ *while the anomaly conditions number 4.* $N_w = 2$ *is the* unique *value where the free parameters equal 1 (overall normalization), making all hypercharge ratios fully determined. For* $N_w \geq 3$ *there are* ≥ 2 *free parameters (underdetermined). Thus* $N_w = 2$ *is uniquely selected, giving 15 fermions.*

Theorem 14 (SU(3) from distinctness and minimality)**.** *The fermion count for color group* $SU(N_c)$ *is* $4N_c + 3$*. The minimum is at* $N_c = 2$ *(11 fermions), but* $N_c = 2$ *gives* $G_c = \mathrm{SU}(2) \cong G'$*, violating distinctness (Theorem 11). The next minimum at* $N_c = 3$ *gives 15 fermions and* $G_c = \mathrm{SU}(3) \ncong \mathrm{SU}(2)$*. SU(3) is uniquely selected.*

Theorem 15 (Universal cubic factorization)**.** *For any* $SU(N_c) \times SU(2) \times U(1)$*, the cubic anomaly condition factors as:*

$$6\, N_c\, (N_c - r - 1)(N_c + r + 1) = 0,$$

giving hypercharges $Y_Q \cdot (1, -(N_c + 1), N_c - 1, -N_c, 2N_c)$*. At* $N_c = 3$*:* $(1, -4, 2, -3, 6) \cdot y_Q$ *— exactly the Standard Model. The factorization is universal; the SM charges are the unique solution of a structural equation, not an accident of 3 colors.*

Theorem 16 (Minimal coupling)**.** $z(\gamma) = e^{i(k\varphi(\gamma) + q \oint A)}$*: propagation* $\times$ *holonomy. Aharonov–Bohm follows.*

4 Quantum and Matter Structure

Theorem 17 (Charge algebra). *The source functional φ is a linear map $V \to \mathbb{R}$. Linearity gives additive charges: $\varphi(d_1 \oplus d_2) = \varphi(d_1) + \varphi(d_2)$; negation gives conjugation: $\varphi(-d) = -\varphi(d)$.*

Theorem 18 (Annihilation). *A defect–antidefect pair $d \oplus \bar{d}$ carries zero net charge: $\varphi(d \oplus (-d)) = 0$. Charge conservation follows from linearity.*

Theorem 19 (Multiple charges). *Given k independent linear functionals $\varphi_1, \ldots, \varphi_k$, each defect carries k independent charges. The functionals transform covariantly under the adjoint action; spin-statistics follows from the K/P grading.*

Theorem 20 (Born rule). *$|z|^2$ is the unique rotation-invariant quadratic observable on the complex amplitude $z = Q + iP$.*

Theorem 21 (Complex amplitudes). *The Frobenius theorem restricts composition algebras over $\mathbb{R}$ to $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$, $\mathbb{O}$. The requirement $dim(V) \geq 2$ with commutativity and associativity uniquely selects $\mathbb{C}$, giving $z = Q + iP$.*

Theorem 22 (Propagation rule). *$z(\gamma) = e^{ik\varphi(\gamma)}$ from linearity $\to$ additivity $\to$ exponential (existence + uniqueness verified). Interference: $|z_1 + z_2|^2 = 2 + 2\cos(k(s_1 - s_2))$.*

Theorem 23 (Decoherence and classical limit). *$\gamma(t) = e^{-\Gamma t}$; classical limit proven (ε-δ).*

5 Symmetry Breaking, CP Violation, and Strong CP

Theorem 24 (Higgs potential). *$V = -a|z|^2 + b|z|^4$: unique quartic rotation-invariant. Vacuum $v^2 = a/(2b)$, $m_h^2 = 4a$, $m_\pi^2 = 0$.*

Theorem 25 (Symmetry breaking from decoherence). *Lindblad dephasing selects a K-sector vacuum, breaking the dressing U(1). P-sector fluctuations = Goldstone mode. No ad hoc potential required.*

Theorem 26 (SM hypercharges). *Given the derived gauge group and representation content (Theorems 13–14), anomaly cancellation uniquely determines $(y_Q, y_u, y_d, y_L, y_e) = y_Q \cdot (1, -4, 2, -3, 6)$.*

Theorem 27 (CP violation [14]). *$(d-1)(d-2)/2$ phases require $d \geq 3$. $N_g \geq 3$ for baryogenesis.*

Theorem 28 (Three generations from the gauge orbit fiber). *The source functional φ is linear in $z \in \mathbb{C}^{N_c}$, so $Q = \mathrm{Re}(z_1)$ is not gauge-invariant: φ depends on the direction in the fiber, not just on gauge orbits. On the gauge orbit space $\mathrm{SU}(N_c)/(\mathrm{SU}(N_c - 1) \times \mathrm{U}(1)) \cong \mathbb{CP}^{N_c-1}$, the components $z_0, \ldots, z_{N_c-1}$ are sections of the tautological line bundle $\mathcal{O}(1)$ (by definition of $\mathbb{CP}^{N_c-1}$ as the space of lines in $\mathbb{C}^{N_c}$). The space of independent holomorphic sections is*

$$\dim H^0(\mathbb{CP}^{N_c-1}, \mathcal{O}(1)) = N_c.$$

Each independent section inherits the same K/P structure and gauge quantum numbers as φ itself, yielding N_c independent four-dimensional replicas of the complete fermion content. For $N_c = 3$ (Theorem 14): $N_g = 3$ exactly.

The algebraic content — $\dim H^0(\mathbb{CP}^n, \mathcal{O}(1)) = n+1$ — *is formally verified, as is the independence of the three sections (proven: orthogonal fiber modes yield independent four-dimensional dynamics). The identification of* $z \in \mathbb{C}^{N_c}$ *with a section of* $\mathcal{O}(1)$ *is not a physical hypothesis — it is the* definition *of the tautological line bundle on* $\mathbb{CP}^{N_c-1}$ *(the space of lines in* $\mathbb{C}^{N_c}$*, whose dual bundle* $\mathcal{O}(1)$ *has the coordinate functions as sections) [18]. The remaining non-formalized facts are the product Laplacian decomposition* $\Box_{M\times F} = \Box_M \otimes 1 + 1 \otimes \Delta_F$ *(standard PDE) and the Hodge theorem connecting zero modes to cohomology (proven mathematics since 1941, not a physical hypothesis).*

Theorem 29 (Strong CP)**.** $\mathrm{Tr}[F\tilde{F}]$ *parity-odd,* φ *parity-even, orthogonal sectors. No axions needed.*

6 Computational Realizations

Observable	Measured	Expected (GR)
Inverse-square α	2.004 ± 0.063	2.0
Shapiro delay slope	3.94	4.0
Deflection	$\propto b^{-1.030}$	b^{-1}

Table 1: Gravitational observables from the causal diamond model.

Benchmark	Result	Accuracy
Double-slit	$d \sin\theta = n\lambda$	0.1%
Bell/CHSH	$\lvert S\rvert = 2\sqrt{2}$	Exact
GHZ/Mermin	$\lvert M\rvert = 2^{N-1}$	Exact

Table 2: Quantum benchmarks. e^{ikL} derived; k sole free parameter.

7 Quantitative Test: Coupling Constants

Coupling	Predicted	Measured	Status
$\alpha_2(M_Z)$	0.036	0.034	7% (zero-parameter)
$\alpha_s(M_Z)$	—	0.118	Non-perturbative
$\alpha_1(M_Z)$	—	0.010	Wrong sign from $g^2 = 1$

The boundary condition $g_2^2 = 1$ at M_P with one-loop running gives $\alpha_2(M_Z)$ to 7%. The same boundary condition applied to SU(3) and U(1) fails: SU(3) enters the strong-coupling regime where one-loop perturbation theory is unreliable, and U(1) produces the wrong sign, indicating that the naive $g^2 = 1$ normalization does not apply uniformly to all gauge factors. The SU(2) prediction stands as a genuine zero-parameter test; the SU(3) and U(1) couplings require non-perturbative methods or a more refined boundary condition.

8 Derivation Integrity

Formally proven (Lean 4, zero sorry). Renormalization rigidity. Null-cone (1+1 and n+1). Bianchi identity. Lovelock uniqueness. Ostrogradsky (second-order from stability). Yang–Mills action + Bianchi. Complex representation classification (chiral algebras identified). Two factors from K/P. Representation structure forced (both-multiplet alternatives vector-like, color parity forces singlet count). Chiral $\not\cong$ vector-like as represented algebras (surjectivity argument). Distinctness from chirality ($G_c \not\cong G'$, via chiral/vector-like incompatibility). Charge determinacy (SU(2) uniquely selected: $N_w = 2$ gives 1 free parameter). SU(3) from distinctness + minimality ($4N_c + 3$, $N_c \geq 3$). Fermion count $7N_w + 1$, minimum 15. Universal cubic factorization ($6N_c(N_c - r - 1)(N_c + r + 1) = 0$). Hypercharge uniqueness. Graviton (bridge equation: $\varphi(K(h)) = 0$, 2 polarizations). Minimal coupling. Higgs potential. SSB from decoherence. Charge algebra. Spin-statistics. Born rule. Complex amplitudes (Frobenius). Propagation rule (existence + uniqueness). Interference. Lindblad decoherence + classical limit. Dimension selection ($d = 3$). Primitive necessity (all 5). Variational Einstein. $\Lambda^2 N = 1$. Two-origin gauge. Chirality. Anomaly cancellation. $N_g = 3$ from $\dim H^0(\mathbb{CP}^2, \mathcal{O}(1)) = 3$ (section count, fiber dependence, and orthogonal independence all proven; product Laplacian decomposition and Hodge theorem are standard mathematics). Strong CP parity.

Computationally verified. GR observables (Table 1). Quantum benchmarks (Table 2). Coupling running (§7).

Honestly primitive. The source functional φ is a framework primitive. Energy boundedness is an additional hypothesis yielding: gauge compactness, second-order equations, Einstein $+\Lambda$. Minimality is a selection principle. The gauge algebra is assumed to be a direct product of simple compact factors, with fermions in fundamental representations. The product structure is motivated by the K/P decomposition (Theorem 9) but is not derived from a classification of all possible gauge algebras. The Yang–Mills integration-by-parts is not formalized as a manifold integral. Operational quantum hypotheses are structural, not Hardy/CDP. One free parameter: ρ. The SU(3) bare coupling requires non-perturbative matching. The Higgs vev is not numerically predicted (the hierarchy problem is restated, not solved). N_g is bounded (≥ 3) by CP violation (Theorem 27) and fixed to exactly 3 by the fiber argument (Theorem 28); the latter uses the product Laplacian decomposition and Hodge theorem, which are standard mathematics not yet in Lean/Mathlib. The mass spectrum is not addressed.

9 Discussion

9.1 The second-order condition from stability (Ostrogradsky)

For a non-degenerate higher-derivative Lagrangian, the Hamiltonian is linear in p_1 and therefore unbounded below (proven; see Woodard [11] for review). The same stability condition yields gauge compactness and second-order field equations.

9.2 The gauge group derivation

The derivation of SU(3)×SU(2)×U(1) proceeds in eight steps, each formally verified: (1) chirality from K/P; (2) complex representations required; (3) SU(N) type from classification + minimality; (4) two factors from K/P structure; (5) representation structure

forced by color parity + chirality (both-multiplet alternatives are vector-like); (6) distinctness from chirality (chiral $\not\cong$ vector-like as represented algebras, via surjectivity); (7) SU(2) from charge determinacy ($N_w = 2$ uniquely gives fully determined charges); (8) SU(3) from distinctness + minimality ($N_c \neq 2$, minimal $N_c \geq 3$). The universal cubic factorization then fixes all hypercharges. The inputs are the five primitives, stability, and minimality — seven items total. Everything else follows.

9.3 Limitations

$N_g = 3$ is derived from the gauge orbit fiber (Theorem 28), with the product Laplacian decomposition and Hodge theorem as the non-formalized mathematical steps. The mass spectrum requires the hierarchy problem. α_s requires Poisson-lattice matching. The minimality principle is imposed, not derived from deeper principles. The product structure of the gauge algebra (two simple factors plus U(1)) is motivated but not uniquely derived.

9.4 Open questions

Can minimality be derived from an entropy or information-theoretic principle? Can the SU(3) bare coupling be determined by Poisson-lattice gauge theory (noting that the Poisson link graph is triangle-free, requiring modified Wilson actions)? What determines the Yukawa couplings and mass hierarchy? Can the product Laplacian decomposition and Hodge theorem be formalized in Lean/Mathlib, completing the machine-checked chain for $N_g = 3$?

10 Conclusion

Seven inputs — five structural primitives, energy boundedness, and minimality — yield the Standard Model gauge group SU(3)×SU(2)×U(1), its complete fermion content, all hypercharge ratios, the Einstein field equation, the Yang–Mills equation, the propagation rule, the Born observable, chirality, the Higgs potential, symmetry breaking, the graviton, minimal coupling, a strong CP resolution, $N_g = 3$ from the gauge orbit fiber, and $\alpha_2(M_Z)$ to 7%. Every algebraic step is formally verified in Lean 4 with zero `sorry`; the generation count additionally uses two standard mathematical results not yet in Lean/Mathlib (product Laplacian decomposition and the Hodge theorem). The remaining open problems — masses, the hierarchy, the SU(3) bare coupling — are sharply identified with concrete paths toward resolution.

Summary of the derivation chain:

Input		Output
φ (source functional)	$\to$	K/P $\to$ chirality $\to$ SU(N) $\to$ two factors $\to$ rep structure $\to$ fiber $\mathcal{O}(1)$ sections $\to N_g = 3$
Lie algebra	$\to$	Yang–Mills
Lorentzian metric	$\to$	Einstein $+\Lambda$
Linear composition	$\to$	Anomaly cancellation $\to$ hypercharges
dim ≥ 2	$\to$	$\mathbb{C}$-amplitudes, e^{ikL}, Born rule
Stability	$\to$	2nd-order, compactness, Higgs
Minimality	$\to$	SU(2), SU(3), 15 fermions

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